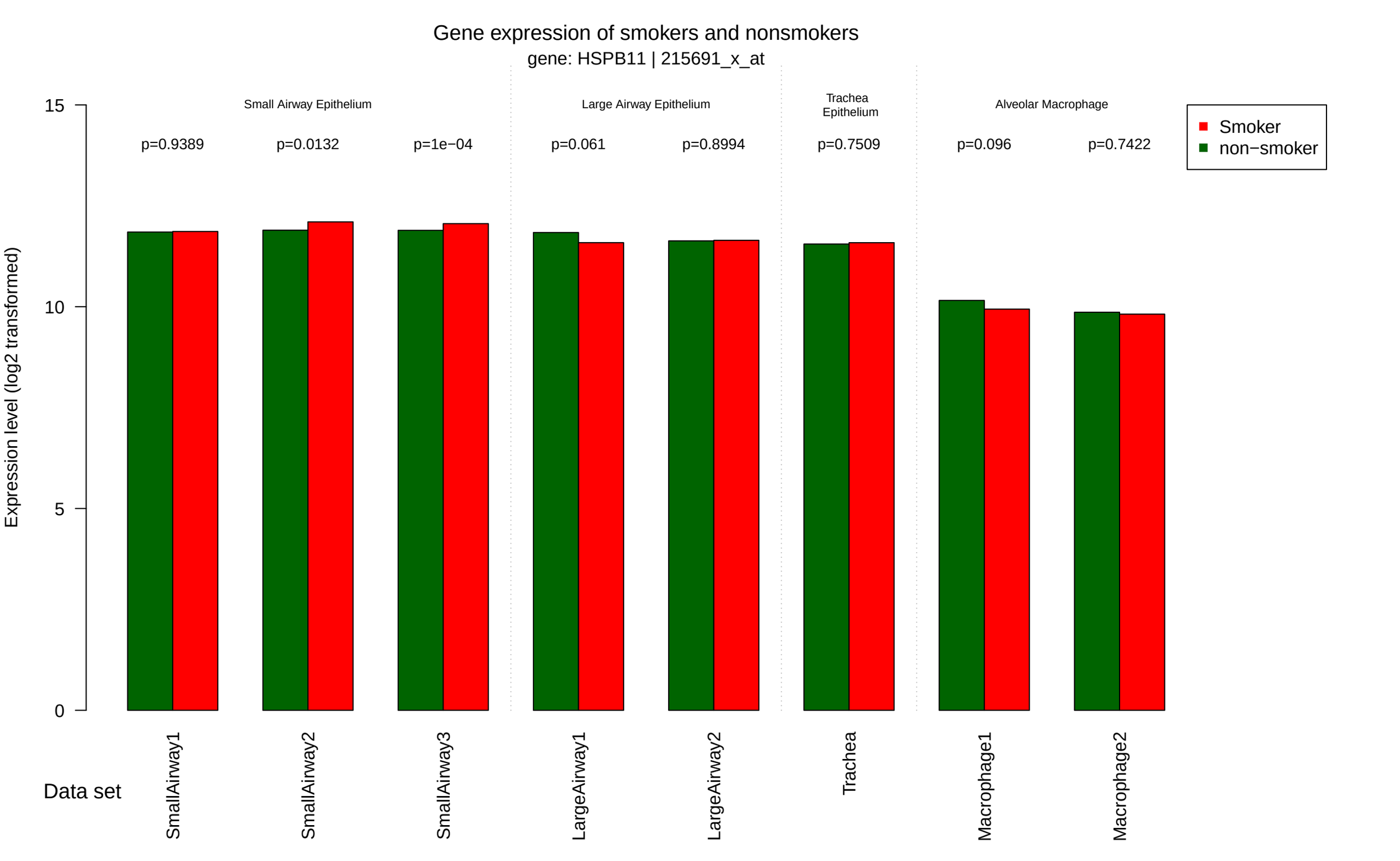
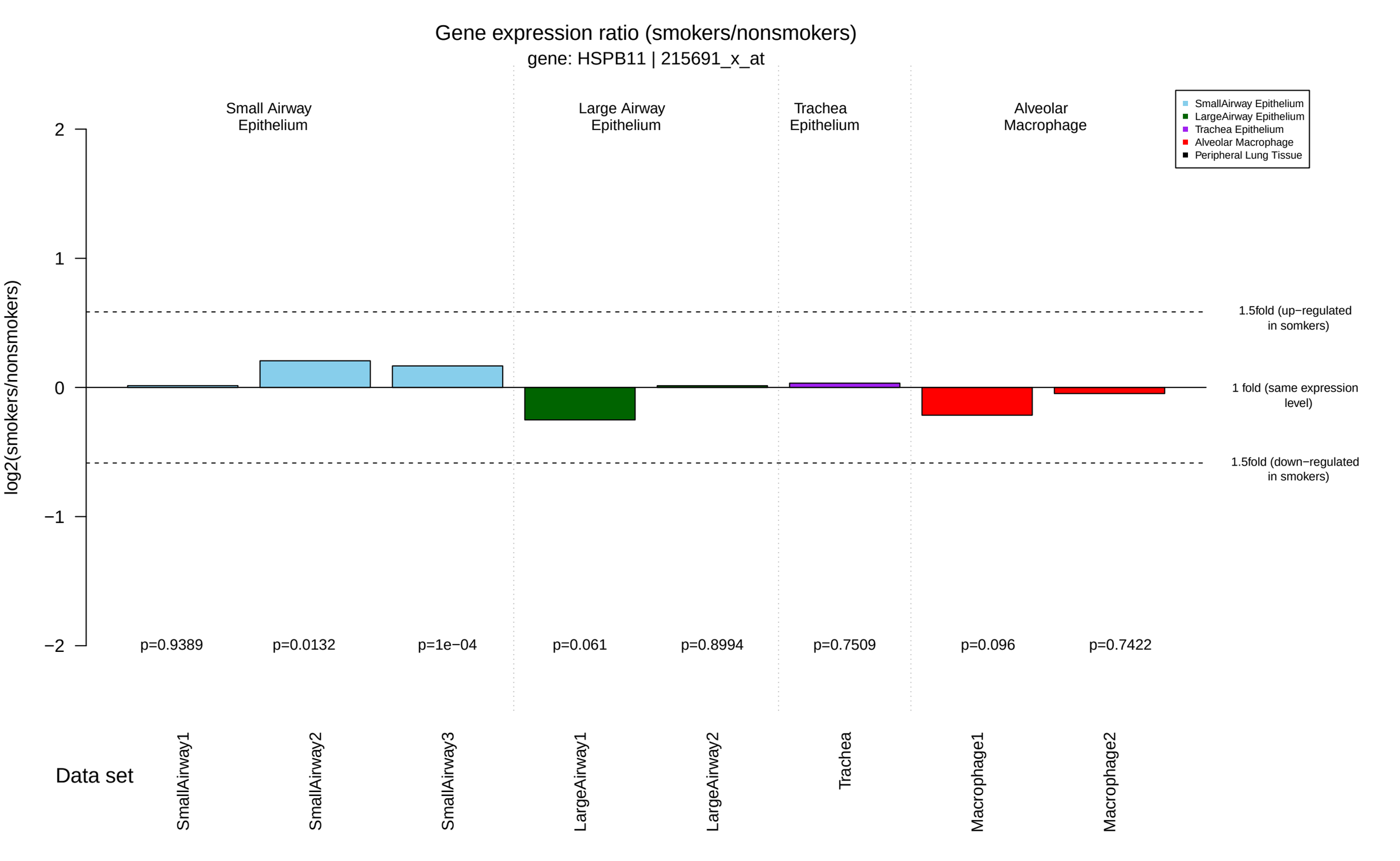
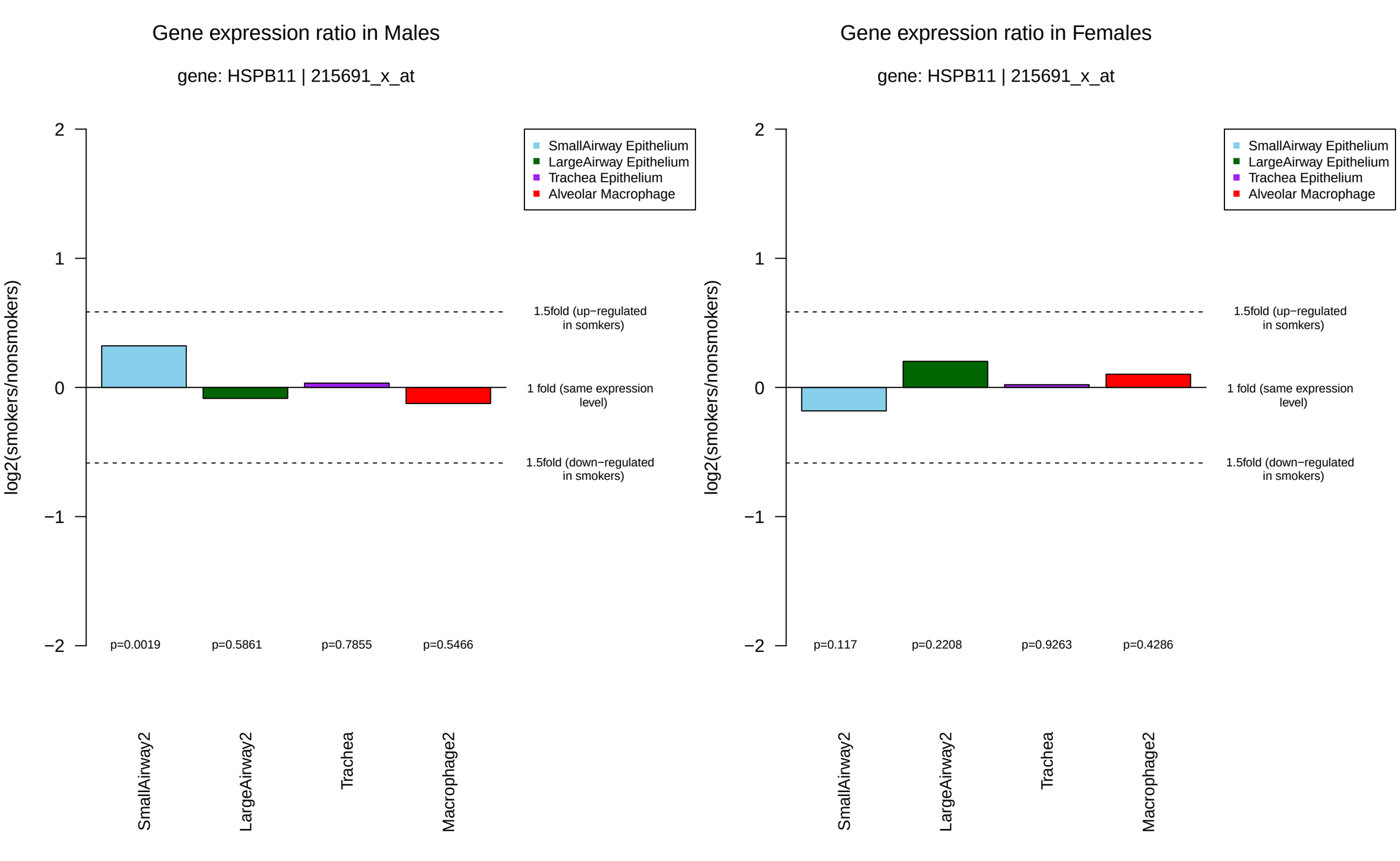
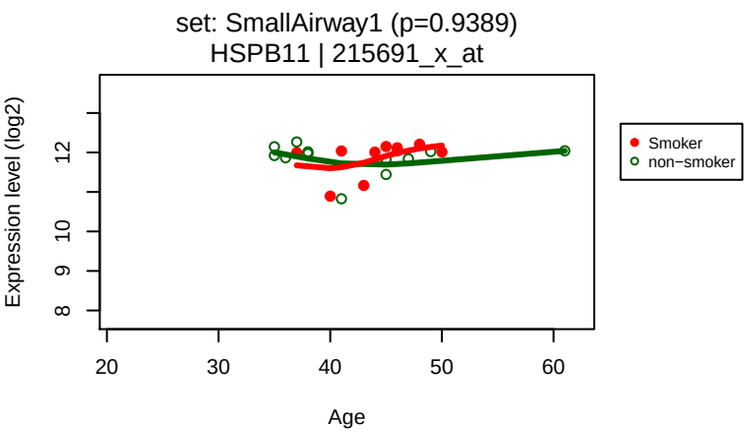
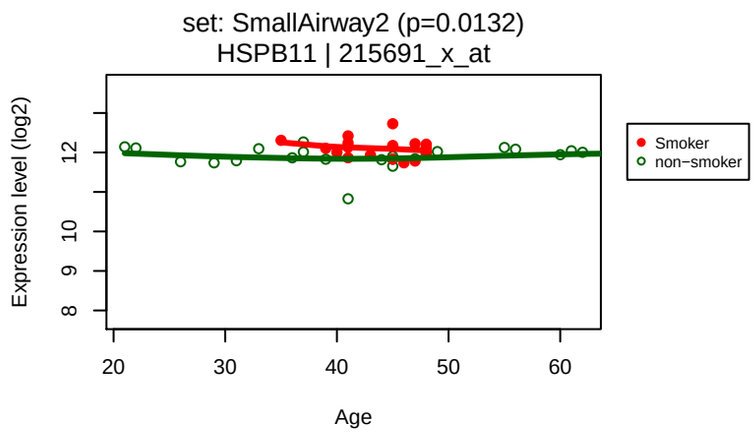
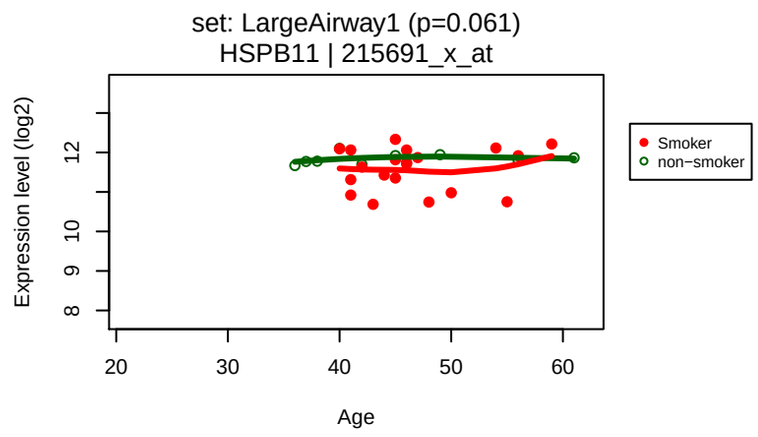

Gene expression of smokers and nonsmokers
gene: HSPB11 | 215691_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9389
p=0.0132
p=1e−04
p=0.061
p=0.8994
p=0.7509
p=0.096
p=0.7422
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

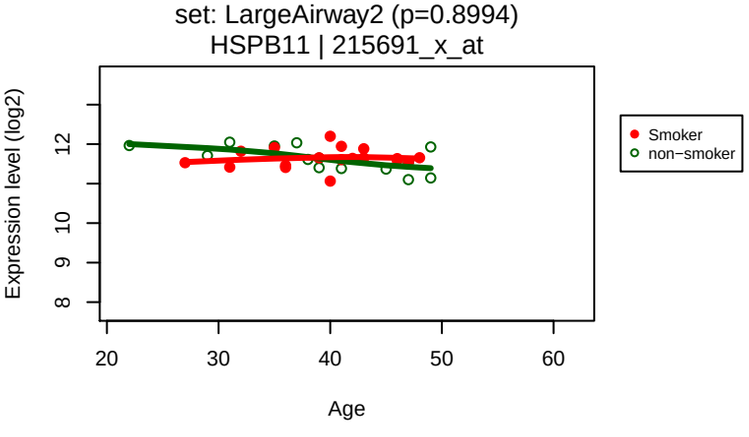

Gene expression ratio (smokers/nonsmokers)
gene: HSPB11 | 215691_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9389
p=0.0132
p=1e−04
p=0.061
p=0.8994
p=0.7509
p=0.096
p=0.7422
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

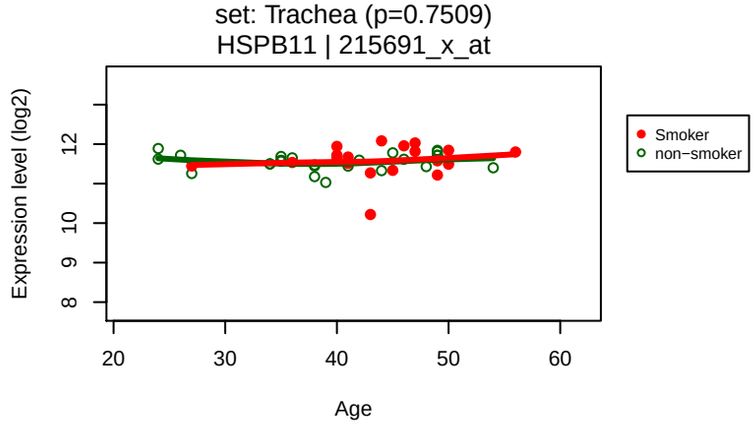

Gene expression ratio in Males
gene: HSPB11 | 215691_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0019
p=0.5861
p=0.7855
p=0.5466
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPB11 | 215691_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.117
p=0.2208
p=0.9263
p=0.4286
SmallAirway2
LargeAirway2
Trachea
Macrophage2

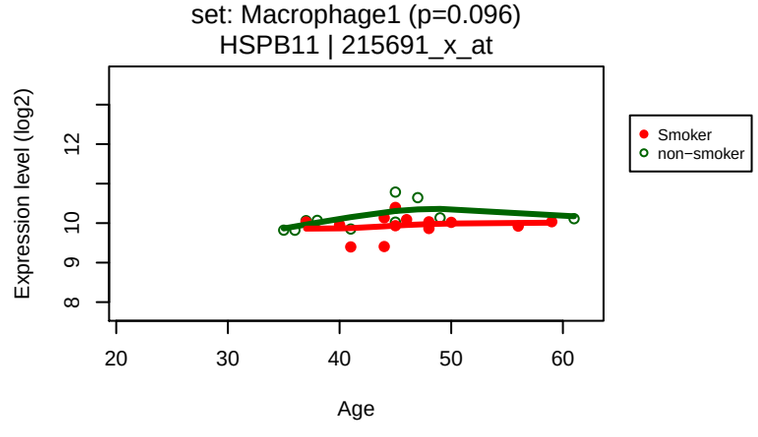

set: SmallAirway1 (p=0.9389)
HSPB11 | 215691_x_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker

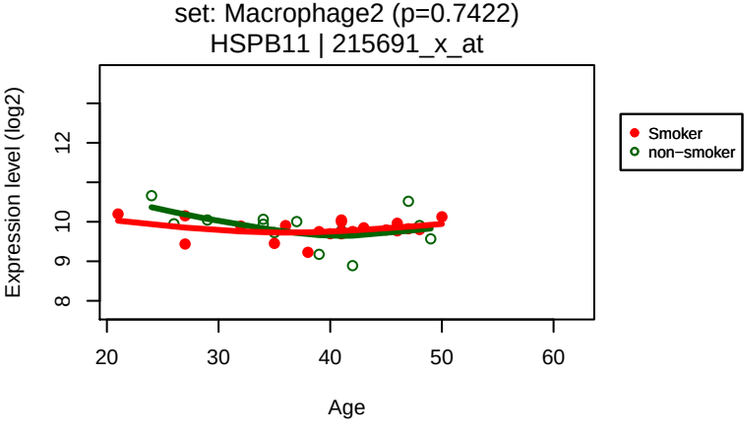

set: SmallAirway2 (p=0.0132)
HSPB11 | 215691_x_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker

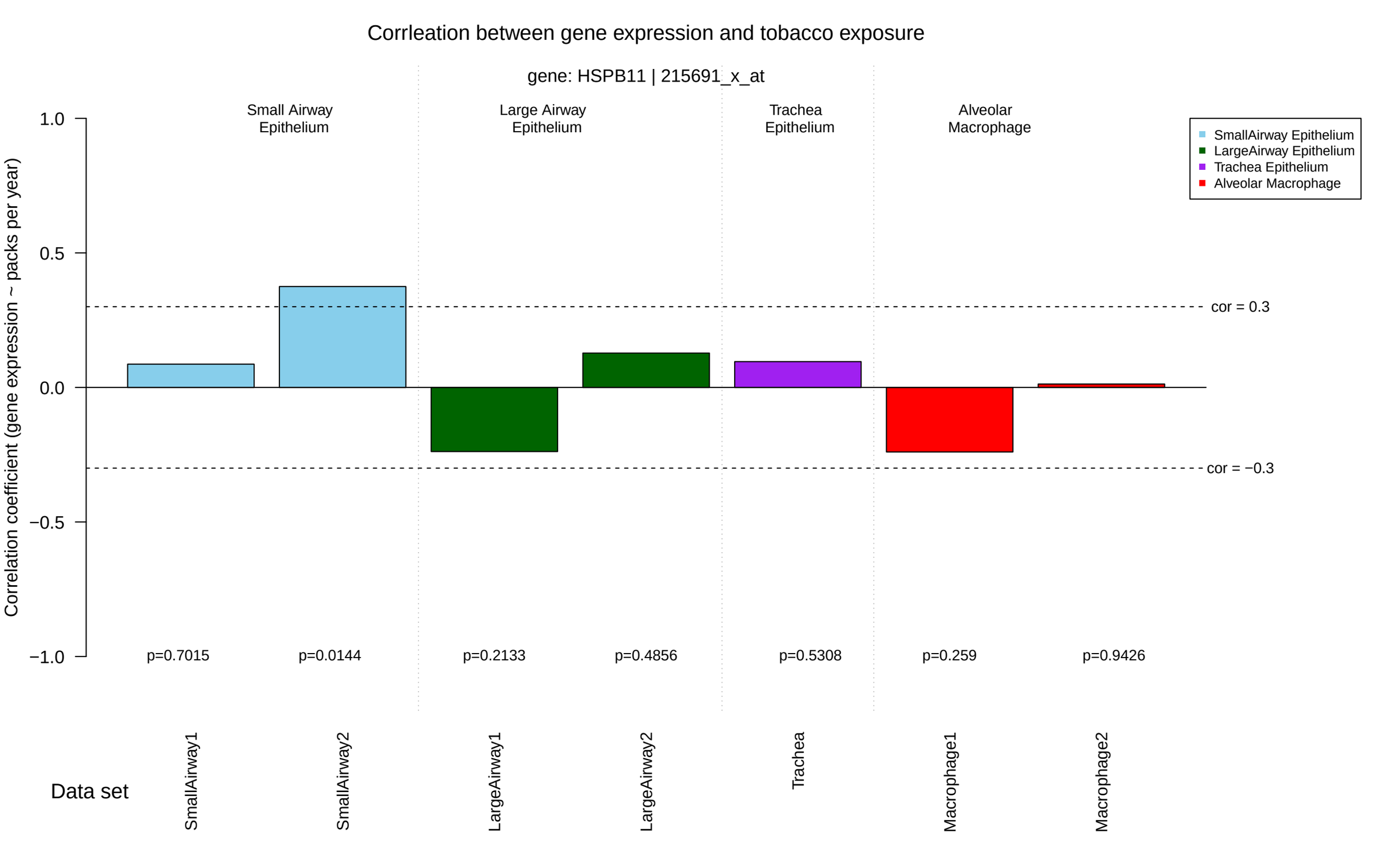

set: LargeAirway1 (p=0.061)
HSPB11 | 215691_x_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8994)
HSPB11 | 215691_x_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.7509)
HSPB11 | 215691_x_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.096)
HSPB11 | 215691_x_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7422)
HSPB11 | 215691_x_at
Expression level (log2)
8
9
10
12
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPB11 | 215691_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7015
p=0.0144
p=0.2133
p=0.4856
p=0.5308
p=0.259
p=0.9426
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2